USN | | | | | | | | | | [22CTISDE512/22CTMADE512]

FIFTH SEMESTER B.Tech EXAMINATION, DECEMBER-2024

ESSENTIALS OF BIG DATA ANALYTICS

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART - A			**5x2=10**
1.	**a.**	Difference between Big data and thick data.	L2	CO1	**02**
	b.	What is the default block size in HDFS?	L1	CO2	**02**
	c.	Mention two advantages of MapReduce.	L1	CO3	**02**
	d.	List two features of Hive.	L1	CO4	**02**
	e.	List any two challenges of Big Data.	L2	CO1	**02**
		PART - B			**6 X 5=30**
2.	**a.**	Discuss the characteristics of Big Data, emphasizing its "5 Vs".	L2	CO1	**05**
		OR			
	b.	What is Apache Hadoop? Discuss its key components.	L2	CO1	**05**
3.	**a.**	In what ways does Hadoop process both structured and unstructured data?	L2	CO4	**05**
		OR			
	b.	What are the primary security challenges in Big Data cloud computing?	L2	CO4	**05**
4.	**a.**	Define Hadoop Distributed File System (HDFS) and explain its role in managing big data.	L2	CO2	**05**
		OR			
	b.	Discuss the key features that make HDFS fault-tolerant and scalable.	L2	CO2	**05**
5.	**a.**	What is MapReduce. Mention best alternatives of MapReduce.	L1	CO3	**05**
		OR			
	b.	What is the role of the Job Submitter in MapReduce?	L1	CO3	**05**
6.	**a.**	Mention the limitations of Hive.	L2	CO4	**05**
		OR			
	b.	What are the key features of HBase?	L2	CO4	**05**
7.	**a.**	Explain the types of machine learning algorithms.	L2	CO5	**05**
		OR			
	b.	What are the key elements required for building good machine learning systems?	L2	CO5	**05**

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		PART - C			6 X 10=60
8.	a.	Discuss in detail the various types of digital data and their implications for modern businesses.	L3	CO1	10
		OR			
	b.	Describe the key cryptographic techniques used in cloud computing. How do they ensure data security and privacy?	L3	CO1	10
9.	a.	Explain the concept of Big Data storage. What are the key concerns related to data security and confidentiality in cloud computing?	L3	CO6	10
		OR			
	b.	Describe IBM's Big Data strategy and how it uses Hadoop to manage and analyze large datasets.	L3	CO6	10
10.	a.	Discuss the different compression and serialization techniques used in Hadoop to manage large volumes of data.	L3	CO4	10
		OR			
	b.	Difference between Apache Sqoop and Apache Flume.	L3	CO2	10
11.	a.	Advantages and Simple tips on how to improve MapReduce performance.	L4	CO3	10
		OR			
	b.	Describe the lifecycle of a MapReduce job from submission to completion.	L4	CO3	10
12.	a.	Discuss the features of Big SQL and its role in the Hadoop ecosystem.	L4	CO4	10
		OR			
	b.	Explain the features of Apache Pig and its advantages over MapReduce.	L4	CO4	10
13.	a.	Explain the steps involved in conducting Data Analysis using R, starting from problem identification to the final visualization stage.	L5	CO5	10
		OR			
	b.	Explain how Big Data and Machine Learning are integrated in the real world. Provide examples from industries using both technologies.	L5	CO5	10